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Whose English is it anyway?: Enhancing language models for text classification tasks

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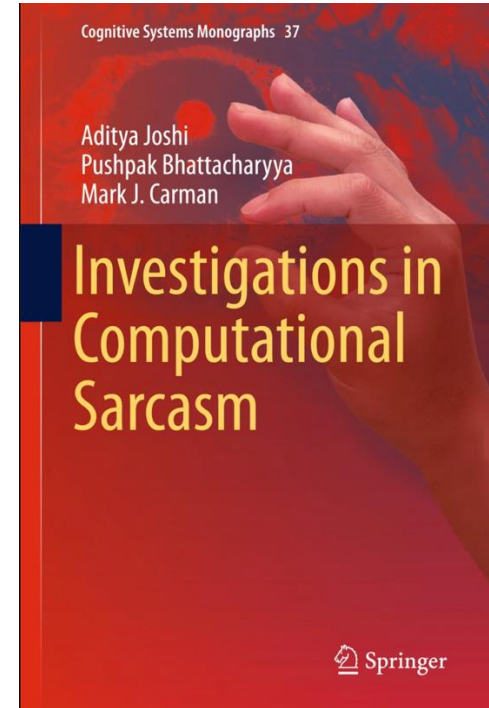
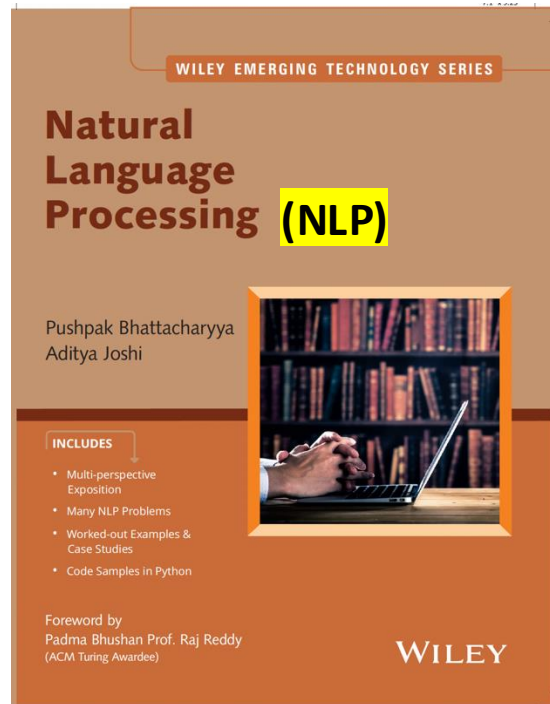


An Acknowledgement of Country is a statement to show respect for Aboriginal and Torres Strait Islander peoples, their cultures, and their ongoing connection to the land.

About me

My CFILT/IITB connection ☺ :

- Masters Project (2010): Sentiment analysis of tweets, IIT Bombay
- PhD Thesis (2018): Investigations in Computational Sarcasm, IIT Bombay & Monash University



University of New South Wales (UNSW)

Group of 8 (Go8) University

Ranked **20th** in the 2026 QS World
University Rankings

~60,000 student enrolments

~20,000 international students from 137 countries

>\$450M research income



CSE at UNSW



1st in Australia

Algorithms, Artificial Intelligence, Databases,
Embedded & Real-Time Systems and Operating
Systems - CS Rankings



1st in Australia

Engineering & Technology Faculty – ARWU
Shanghai Subject Rankings and QS World University
Rankings by Subject



1st in Australia

Teaching and Research – Times Higher Education
University Rankings

Number of Students:

Undergraduate: 10,372

Postgraduate (Coursework): 6000

Postgraduate (Research): 281

Number of Staff:

Academic: 75

Professional / Technical / Admin Staff: 29



NLP Research Group @ UNSW

- 2 postdocs
- 3 affiliated faculty members
- 7 PhD students
- 3
- 3 Masters students
- 5 Bachelors students



Welcome to the UNSW NLP Group!

<https://unswnlp.github.io/>

School Homepage: <https://www.unsw.edu.au/engineering/our-schools/computer-science-and-engineering/our-research/research-groups/natural-language-processing>

Breaking News!!!!

So I asked GPT-5 if “PhD-level” intelligence is true or just marketing:



OpenAI claims GPT-5 model boosts ChatGPT to 'PhD level'

4 hours ago

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Lily Jamali
North America Technology correspondent

Liv McMahon
Technology reporter



<https://www.bbc.com/news/articles/cy5prvgw0r1o>

" What's Marketing Spin:

PhD-Level ≠ General Intelligence

A PhD thinks critically, asks questions, and knows what it doesn't know. GPT models don't.

GPT-5 can imitate PhD-level reasoning in specific contexts (e.g., summarizing papers, solving a math problem), but it lacks true understanding, self-awareness, and original insight.

“Better than most programmers” has caveats

The model performs well when problems are clearly stated. It struggles with ambiguous specs, changing requirements, or long-term codebase planning — where human judgment, experience, and team communication matter.



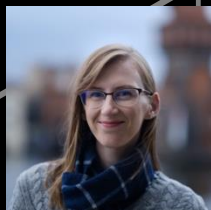
In Short:

Yes, GPT-5 is shockingly capable in coding and academic reasoning.

No, it's not a PhD. It's not magic. It's a very powerful tool — not a thinking being.

"

Transcript citation: <https://chatgpt.com/share/68953ced-8ed8-800c-b691-ae617b696bbf>



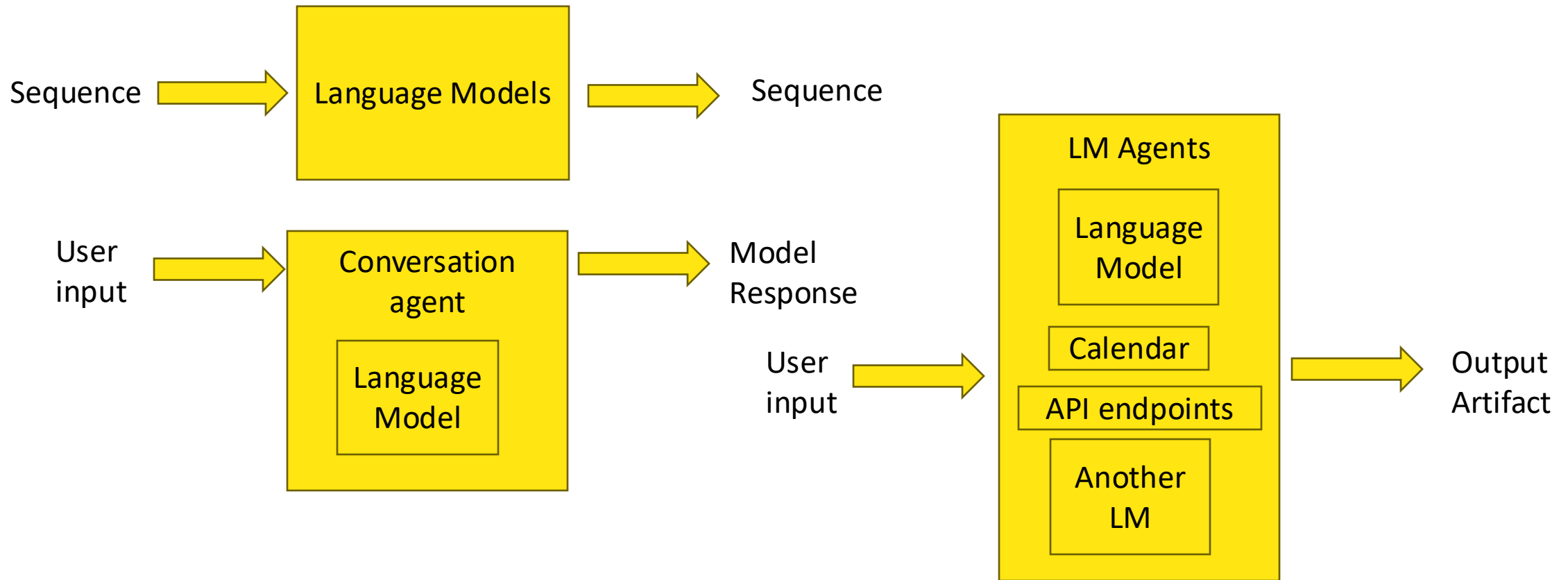
Raj Dabre

Part 1:

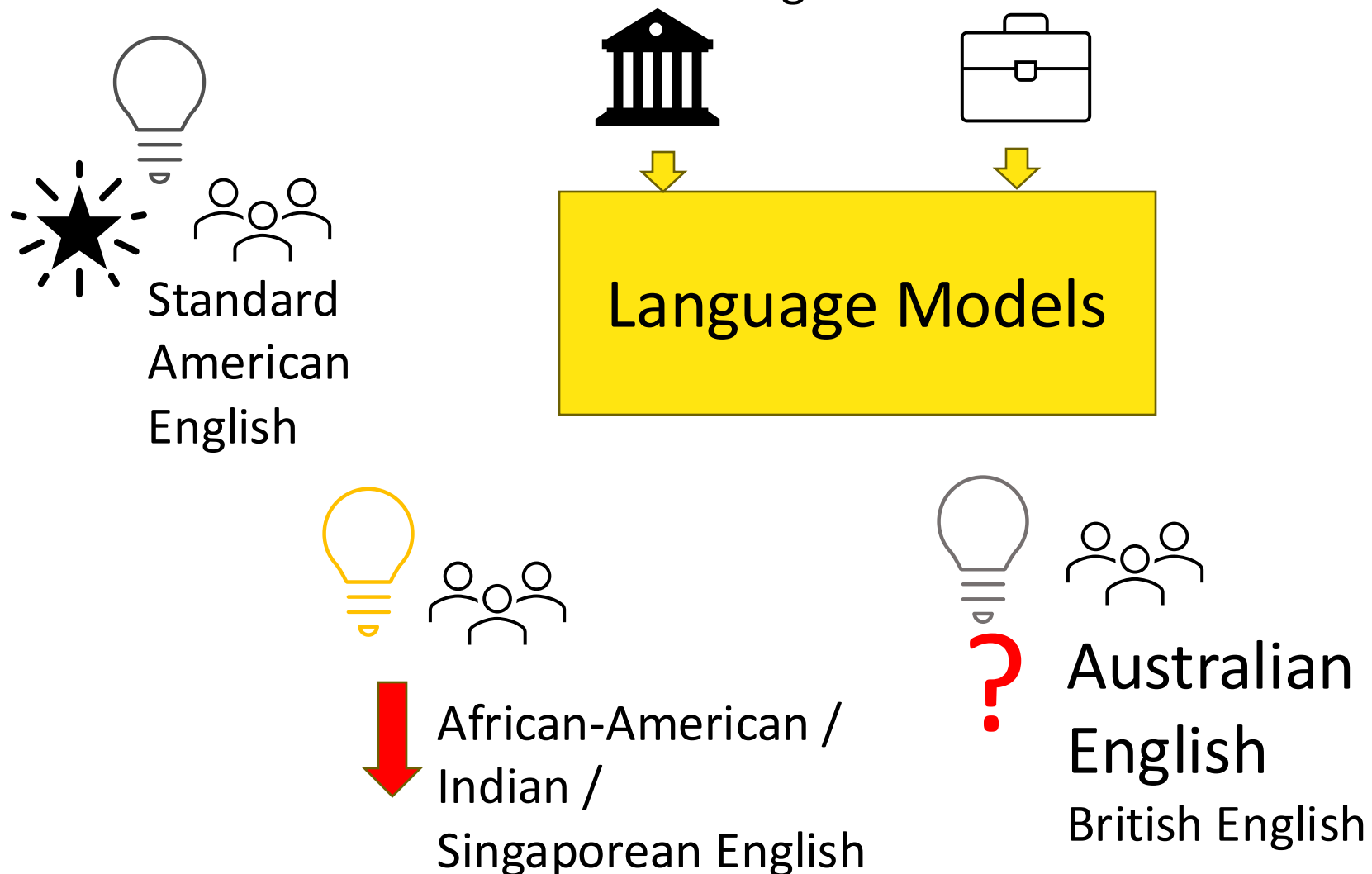
Topic and Sentiment/Sarcasm
Classification for Varieties of English

Partially funded by Google Research

Language Models have ushered new opportunities



.. but these opportunities are not equally accessible for all users of English.



English



Countries or territories where English is an official language [\[edit source \]](#)

Main article: [List of countries and territories where English is an official language](#)

English is an [official language](#) (*de facto* or *de jure*) of the following countries and territories.^[14]

- **North America:** [Anguilla](#), [Antigua and Barbuda](#), [Bahamas](#), [Barbados](#), [Belize](#), [Bermuda](#), [British Virgin Islands](#), [Canada](#), [Cayman Islands](#), [Curaçao](#), [Dominica](#), [Grenada](#), [Jamaica](#), [Montserrat](#), [Puerto Rico](#), [Saint Kitts and Nevis](#), [Saint Lucia](#), [Saint Vincent and the Grenadines](#), [Trinidad and Tobago](#), [Turks and Caicos Islands](#), [United States](#), [United States Virgin Islands](#)
- **South America:** [Falkland Islands](#), [Guyana](#)
- **Europe:** [Akrotiri and Dhekelia](#), [Gibraltar](#), [Guernsey](#), [Ireland](#), [Isle of Man](#), [Jersey](#), [Malta](#), [United Kingdom](#)
- **Africa:** [Botswana](#), [British Indian Ocean Territory](#), [Cameroon](#), [Eswatini](#), [Gambia](#), [Ghana](#), [Kenya](#), [Lesotho](#), [Liberia](#), [Malawi](#), [Mauritius](#), [Namibia](#), [Nigeria](#), [Rwanda](#), [Saint Helena, Ascension and Tristan da Cunha](#), [Sierra Leone](#), [South Africa](#), [South Sudan](#), [Sudan](#), [Tanzania](#), [Uganda](#), [Zambia](#), [Zimbabwe](#)
- **Asia:** [Christmas Island](#), [Cocos \(Keeling\) Islands](#), [Hong Kong](#), [India](#), [Pakistan](#), [Philippines](#), [Singapore](#)
- **Oceania:** [American Samoa](#), [Australia](#), [Cook Islands](#), [Fiji](#), [Guam](#), [Kiribati](#), [Marshall Islands](#), [Micronesia](#), [Nauru](#), [New Zealand](#), [Niue](#), [Norfolk Island](#), [Northern Mariana Islands](#), [Palau](#), [Papua New Guinea](#), [Pitcairn Islands](#), [Samoa](#), [Solomon Islands](#), [Tokelau](#), [Tuvalu](#), [Vanuatu](#)

Although not official, English is also an important language in some former colonies and [protectorates](#) of the [British Empire](#) where it is used as an [administrative language](#), namely [Brunei](#), [Malaysia](#), and [Sri Lanka](#).

Language Varieties

Variety of a language that is a characteristic of a particular group of the language's speakers

National Variety: Country of origin

Indian English, British English, Irish English

Varieties encompass orthography, syntax, and vocabulary.

-> What are 'thongs'? What's an 'ambo'?

-> "I am like this only."

-> What does "yeah.. nah" really mean?

Other varieties?

Language Models & Dialectal Variations

Aditya Joshi, Raj Dabre, Diptesh Kanojia, Zhuang Li, Haolan Zhan, Gholamreza Haffari, and Doris Dippold. 2025. Natural Language Processing for Dialects of a Language: A Survey. ACM Comput. Surv. 57, 6, Article 149 (2025), 37 pages. <https://doi.org/10.1145/3712060>

Table 1. Examples of Adverse Impact on NLP Task Performance Due to Dialectal Variations

NLP Task	Paper	Impact
Language classification	[Blodgett et al. 2016]	Language detection shows lower performance for African-American English.
Sentiment classification	[Okpala et al. 2022]	Text in African-American English may be predicted more commonly as hate speech.
Natural Language Understanding	[Ziems et al. 2022]	Popular models perform worse on GLUE tasks for African-American English text.
Summarisation	[Keswani and Celis 2021]	Generated multi-document summaries may be biased towards majority dialect.
Machine translation	[Kantharuban et al. 2023]	Significant drop in MT from and to dialects of Portuguese/Bengali/etc. to and from English.
Parsing	[Scannell 2020]	Lower performance of parsers on Manx Gaelic as compared to Irish/Scottish Gaelic.

For details, tutorial:

Connecting Ideas in ‘*Lower-Resource*’ Scenarios: NLP for National Varieties, Creoles and Other Low-resource Scenarios

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<https://github.com/surrey-nlp/COLING-Tutorial-LowResScene-2025>

Language Models do have a “default”

Variety of English	# Features: Inputs	# Features: Outputs	% Retention ↑
SAE	295	230	78%
SBE	291	210	72%
Indian	73	12	16%
Nigerian	44	5.5	13%
Kenyan	90	9	10%
Irish	26	1	4%
AAE	63	2	3%
Scottish	37	1	3%
Singaporean	40	1	3%
Jamaican	51	1	2%

Table 1: Overview of language varieties and features represented in inputs and GPT-3.5 outputs.

Eve Fleisig, Genevieve Smith, Madeline Bossi, Ishita Rustagi, Xavier Yin, and Dan Klein. 2024. Linguistic Bias in ChatGPT: Language Models Reinforce Dialect Discrimination. In EMNLP.

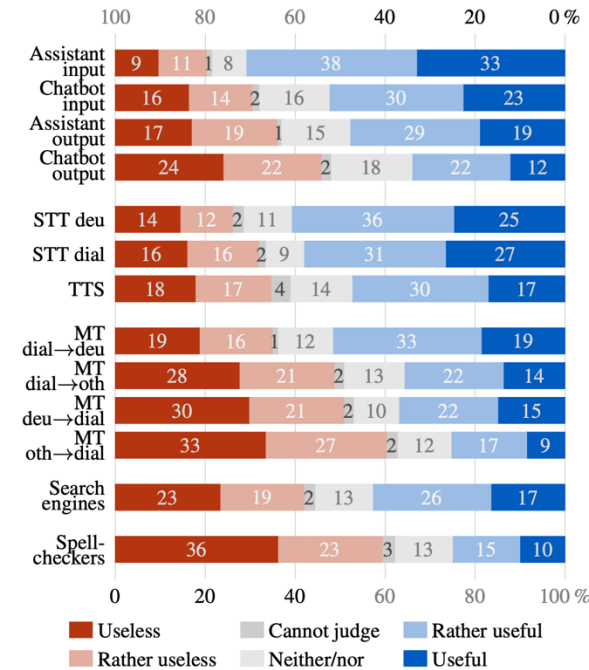


Figure 2: **Opinions on potential language technologies for dialects.** *STT*=speech-to-text, *TTS*=text-to-speech, *dial*=dialect, *deu*=German, *oth*=other languages, *MT*=machine translation, *cannot judge*=skip question.

Verena Blaschke, Christoph Purschke, Hinrich Schuetze, and Barbara Plank. 2024. [What Do Dialect Speakers Want? A Survey of Attitudes Towards Language Technology for German Dialects](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 823–841, Bangkok, Thailand. Association for Computational Linguistics.

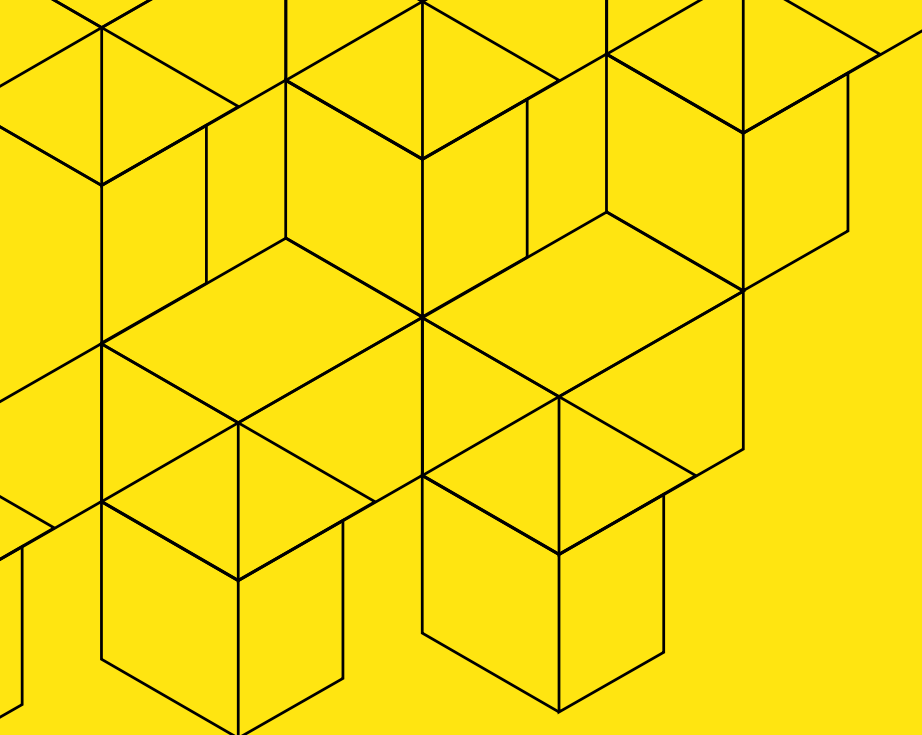
.. and so do existing benchmarks

Task	DEP.	40	3	2	4		3	3		4		3	3			1		3		3					8
	POS.	51	6	2	4		2	3		5		3	3	8		1		3		3					8
	NER	85	2	8	4		4	6	4	5	2	6	3			5	2	2	4		3	3	3		19
	EQA	24	7				11								2								2	2	
	MRC	11	6				1	2																2	
	NLI	38	9	2	2		1	3	3	4		1	2			3	2				1			5	
	TC	38	9	2	2		1	3	3	4		1	2			3	2				1			5	
	SA	9	9																						
	DId	49	26	4			3	4		6	6														
	MT	114	25	23	20	21			8	1	2		3		5		2							4	
	Total	281	42	31	26	21	19	13	12	11	11	8	8	8	6	5	5	4	4	3	3	3	3	3	32
	Total	arabic	high german	italian romance	basque	anglic	sinitic	common turkic	sw shift. romance	greek	gallo-rhaetian	norwegian	neva	bengali	gallo-italian	kurdish	komi	serb.-croa.-bosnian	tupi-guarani.	modern dutch	eastern romance	frisian	swahili	Other	
	Language Clusters																								

Faisal, F., Ahia, O., Srivastava, A., Ahuja, K., Chiang, D., Tsvetkov, Y., & Anastasopoulos, A. (2024). DIALECTBENCH: A NLP Benchmark for Dialects, Varieties, and Closely-Related Languages. *ACL 2024*.

GPT-4 Technical Report. OpenAI.

	GPT-4	GPT-3.5	LM SOTA	SOTA
	Evaluated few-shot	Evaluated few-shot	Best external LM evaluated few-shot	Best external model (incl. benchmark-specific tuning)
MMLU [49]	86.4%	70.0%	70.7%	75.2%
Multiple-choice questions in 57 subjects (professional & academic)	5-shot	5-shot	5-shot U-PaLM [50]	5-shot Flan-PaLM [51]



Whose English is it anyway?

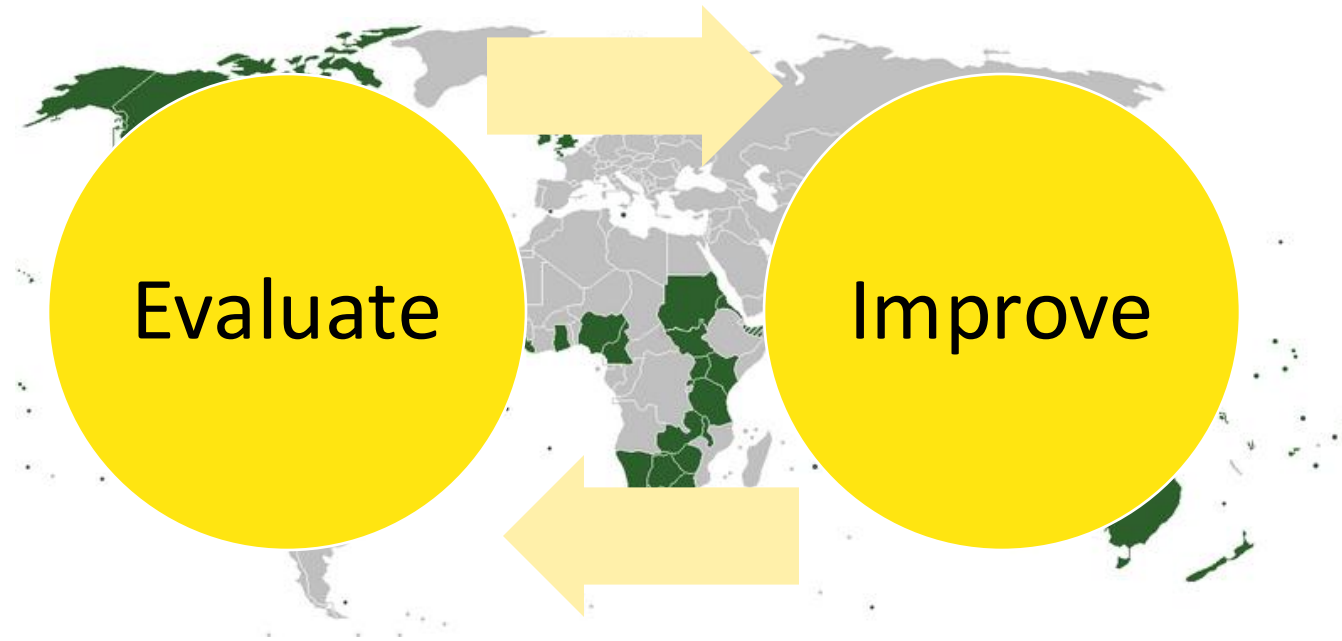


UNSW Strategy: Progress for All

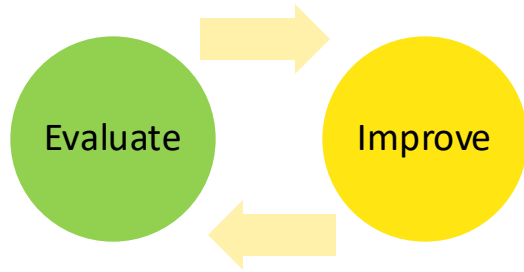
Learn more ↓

<https://www.unsw.edu.au/study/progressforall>

Our Research at UNSW



Part I: Evaluate



Evaluating Dialect Robustness of Language Models via Conversation Understanding

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¹University of New South Wales, Sydney, Australia

²Indian Institute of Technology Bombay, India

{d.srirag, aditya.joshi}@unsw.edu.au nihar@cse.iitb.ac.in

TL;DR: We evaluate how well language models detect topic of conversations .. and they struggle for non-native English.

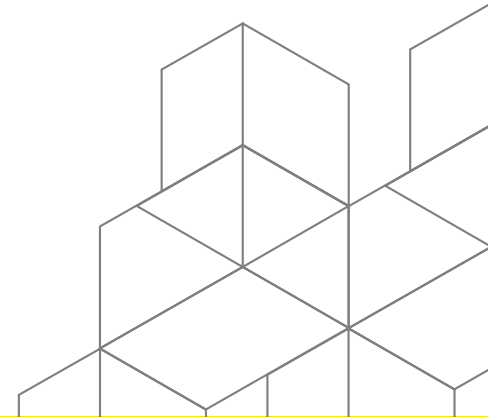
What is a suitable task to evaluate conversation understanding?

Causal language modeling

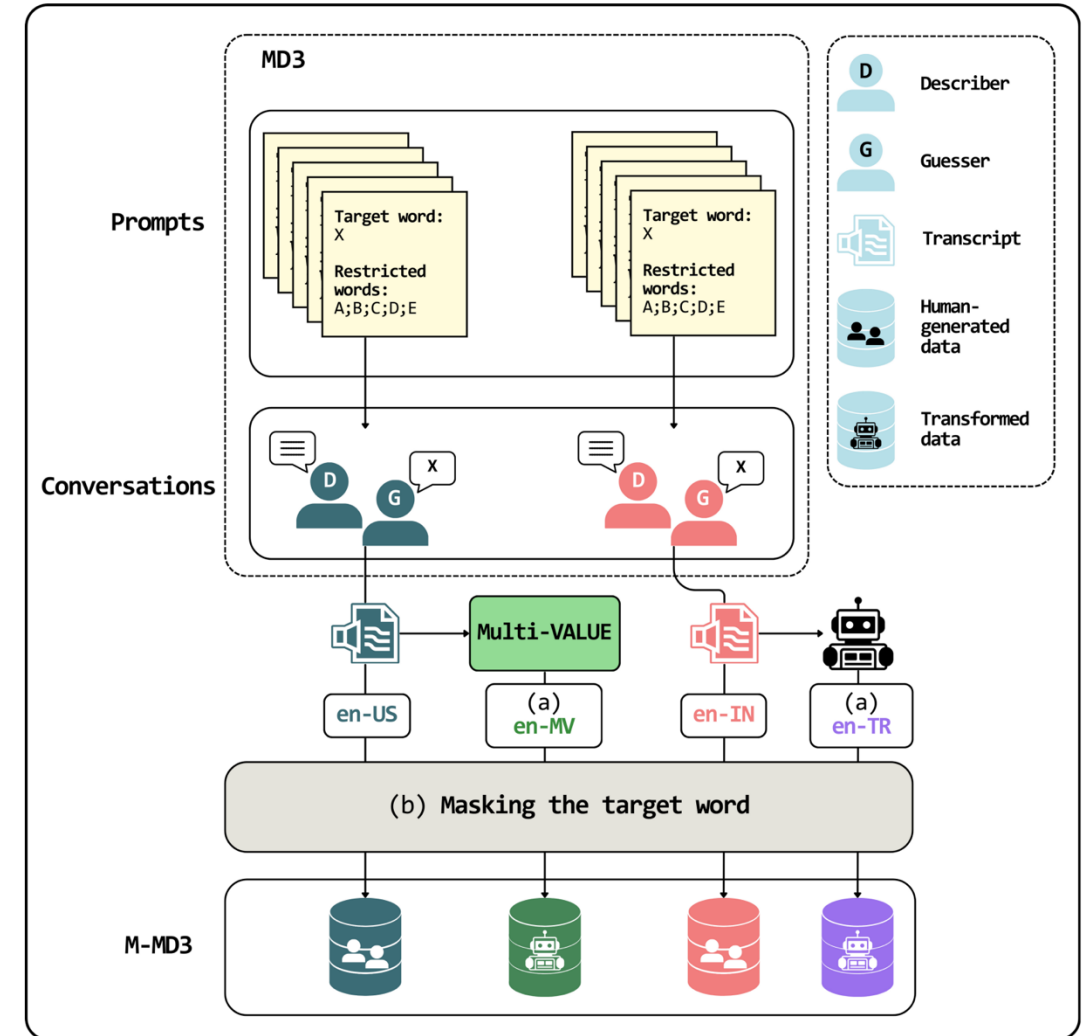
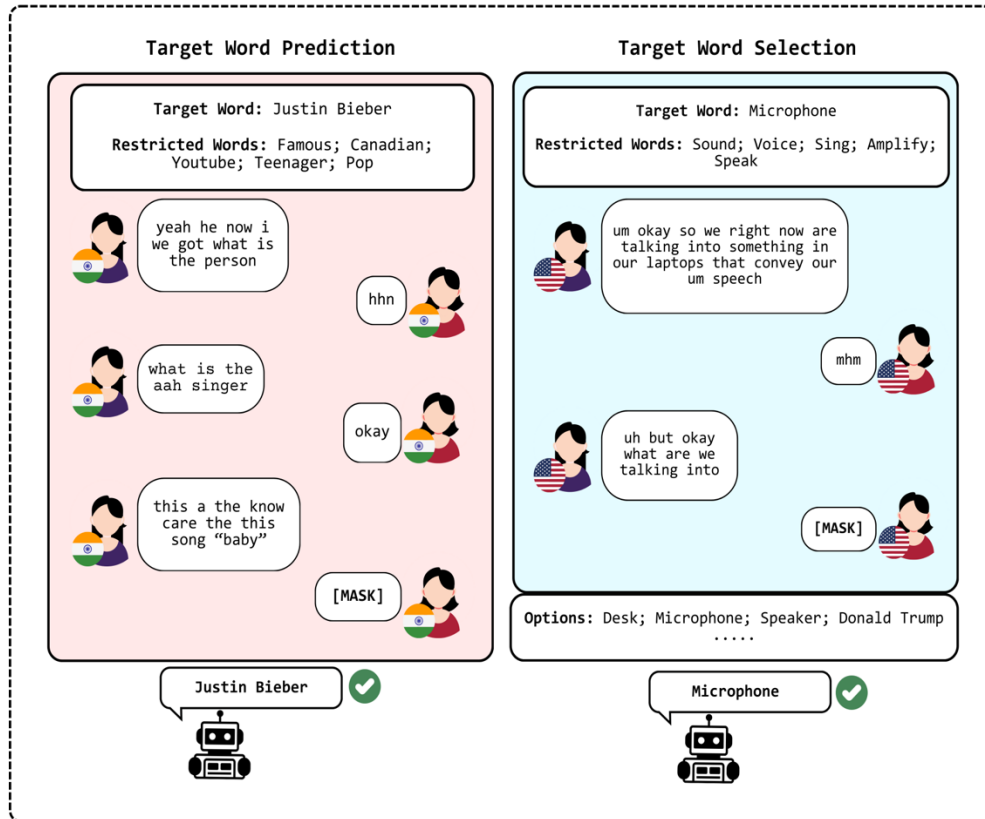
Next token embedding must capture the context of the text so far

How about predicting the topic of the conversation as the next word?

What (human) game involves this?



Dataset & Task



Datasets & Experiment Setup

Subset	Train	Valid	Test
en-US	62	41	311
en-IN	31	21	160
en-MV	49	33	250
en-TR	23	17	131

Table 2: Statistics of M-MD3.

Experiments on GPT-4 and GPT-3.5 are conducted using OpenAI’s API⁹. GPT-3.5 is fine-tuned for 5 epochs, separately for every subset. We select top_p as 0.2 to restrict variability in output generation.

LLAMA-3 is fine-tuned for 20 epochs, with a batch size of 16, Paged 8-bit AdamW (Dettmers et al., 2022) as the optimiser and a learning rate of 2e-4. We use QLoRA adaptors, targeting all linear layers, as recommended by Dettmers et al. (2024). All experiments on LLAMA-3 were performed using two A100 GPUs.

Results

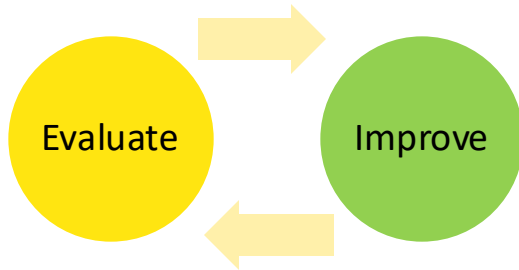
Model	Subset	TWP						TWS					
		Similarity			Accuracy			Similarity			Accuracy		
		PT	FT	Δ	PT	FT	Δ	PT	FT	Δ	PT	FT	Δ
GPT-4	en-US	<u>77.4</u>	-	-	<u>67.8</u>	-	-	<u>85.7</u>	-	-	<u>78.8</u>	-	-
	en-IN	<u>63.0</u>	-	-	<u>45.6</u>	-	-	<u>79.0</u>	-	-	<u>72.5</u>	-	-
	en-MV	<u>75.6</u>	-	-	<u>60.0</u>	-	-	<u>83.6</u>	-	-	<u>74.4</u>	-	-
	en-TR	<u>62.8</u>	-	-	<u>45.8</u>	-	-	<u>83.4</u>	-	-	<u>77.1</u>	-	-
	δ	-14.4	-	-	-22.0	-	-	-6.7	-	-	-6.3	-	-
GPT-3.5	en-US	66.3	72.2	5.9	52.7	59.1	6.4	66.4	80.8	14.4	50.8	71.3	20.5
	en-IN	53.2	59.1	5.9	34.4	40.0	5.6	61.9	70.7	8.8	47.5	60.6	13.1
	en-MV	57.6	71.3	13.7	40.0	54.4	14.4	52.4	71.5	19.1	31.6	57.6	26.0
	en-TR	59.4	<u>61.0</u>	1.6	39.7	41.2	1.5	70.7	73.0	2.3	57.3	60.3	3.0
	δ	-13.1	-13.1	-	-18.3	-19.1	-	-4.5	-10.1	-	-21.0	-16.2	-
LLAMA-3	en-US	70.8	<u>78.0</u>	7.2	60.5	<u>65.3</u>	4.8	78.0	<u>81.8</u>	3.8	67.5	<u>74.6</u>	7.1
	en-IN	59.8	<u>66.3</u>	6.5	43.8	<u>54.4</u>	10.6	68.8	<u>80.8</u>	12.0	56.9	<u>74.4</u>	17.5
	en-MV	68.6	<u>73.8</u>	5.2	54.0	<u>61.6</u>	7.6	72.3	<u>77.6</u>	5.3	58.8	<u>67.2</u>	8.4
	en-TR	60.7	57.5	-3.2	<u>45.8</u>	<u>42.7</u>	-3.1	70.8	<u>79.5</u>	8.7	60.3	<u>72.5</u>	12.2
	δ	-11.0	-11.7	-	-16.7	-10.9	-	-9.2	-1.8	-	-10.6	-0.2	-

Results

Model	Subset	TWP						TWS					
		Similarity			Accuracy			Similarity			Accuracy		
		PT	FT	Δ	PT	FT	Δ	PT	FT	Δ	PT	FT	Δ
GPT-4	en-US	<u>77.4</u>	-	-	<u>67.8</u>	-	-	<u>85.7</u>	-	-	<u>78.8</u>	-	-
	en-IN	<u>63.0</u>	-	-	<u>45.6</u>	-	-	<u>79.0</u>	-	-	<u>72.5</u>	-	-
	en-MV	<u>75.6</u>	-	-	<u>60.0</u>	-	-	<u>83.6</u>	-	-	<u>74.4</u>	-	-
	en-TR	<u>62.8</u>	-	-	<u>45.8</u>	-	-	<u>83.4</u>	-	-	<u>77.1</u>	-	-
	δ	-14.4	-	-	-22.0	-	-	-6.7	-	-	-6.3	-	-
GPT-3.5	en-US	66.3	72.2	5.9	52.7	59.1	6.4	66.4	80.8	14.4	50.8	71.3	20.5
	en-IN	53.2	59.1	5.9	34.4	40.0	5.6	61.9	70.7	8.8	47.5	60.6	13.1
	en-MV	57.6	71.3	13.7	40.0	54.4	14.4	52.4	71.5	19.1	31.6	57.6	26.0
	en-TR	59.4	<u>61.0</u>	1.6	39.7	41.2	1.5	70.7	73.0	2.3	57.3	60.3	3.0
	δ	-13.1	-13.1	-	-18.3	-19.1	-	-4.5	-10.1	-	-21.0	-16.2	-
LLAMA-3	en-US	70.8	<u>78.0</u>	7.2	60.5	<u>65.3</u>	4.8	78.0	<u>81.8</u>	3.8	67.5	<u>74.6</u>	7.1
	en-IN	59.8	<u>66.3</u>	6.5	43.8	<u>54.4</u>	10.6	68.8	<u>80.8</u>	12.0	56.9	<u>74.4</u>	17.5
	en-MV	68.6	<u>73.8</u>	5.2	54.0	<u>61.6</u>	7.6	72.3	<u>77.6</u>	5.3	58.8	<u>67.2</u>	8.4
	en-TR	60.7	57.5	-3.2	<u>45.8</u>	<u>42.7</u>	-3.1	70.8	<u>79.5</u>	8.7	60.3	<u>72.5</u>	12.2
	δ	-11.0	-11.7	-	-16.7	-10.9	-	-9.2	-1.8	-	-10.6	-0.2	-

Interesting discussion on the types of errors in the paper.

Part 2: Improve



Predicting the Target Word of Game-playing Conversations using a Low-Rank Dialect Adapter for Decoder Models

Dipankar Srirag  **Aditya Joshi**  **Jacob Eisenstein** 
 University of New South Wales, Sydney  Google DeepMind
{d.srirag, aditya.joshi}@unsw.edu.au jeisenstein@google.com

TL;DR: We present a method for dialect adaptation of language models.

Task Definition

Target Word Prediction:

Given a masked dialogue between dialectal speakers playing taboo game, predict the target word.

Dataset:

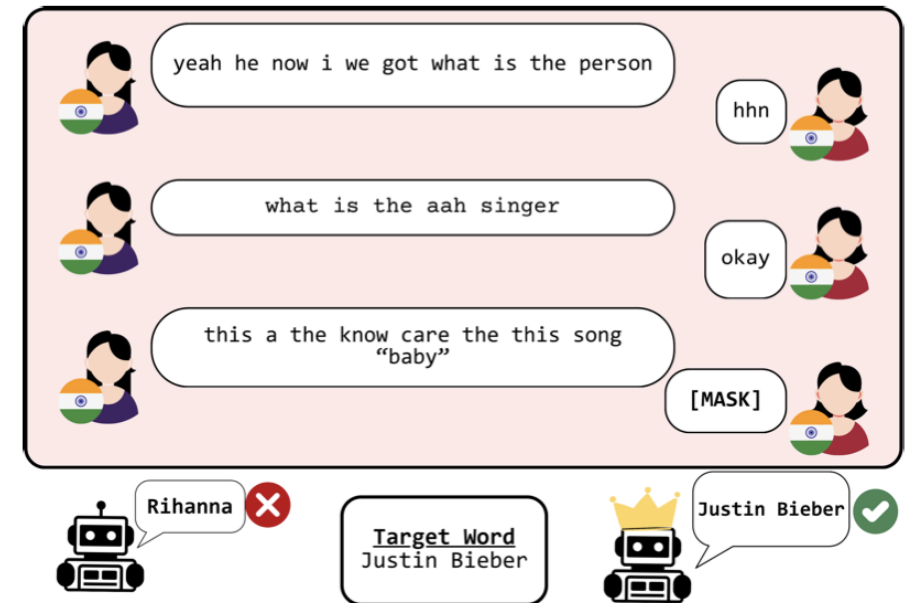
MD-3 ([Eisenstein et al., 2023](#)).

Dialects:

American English(en-US)

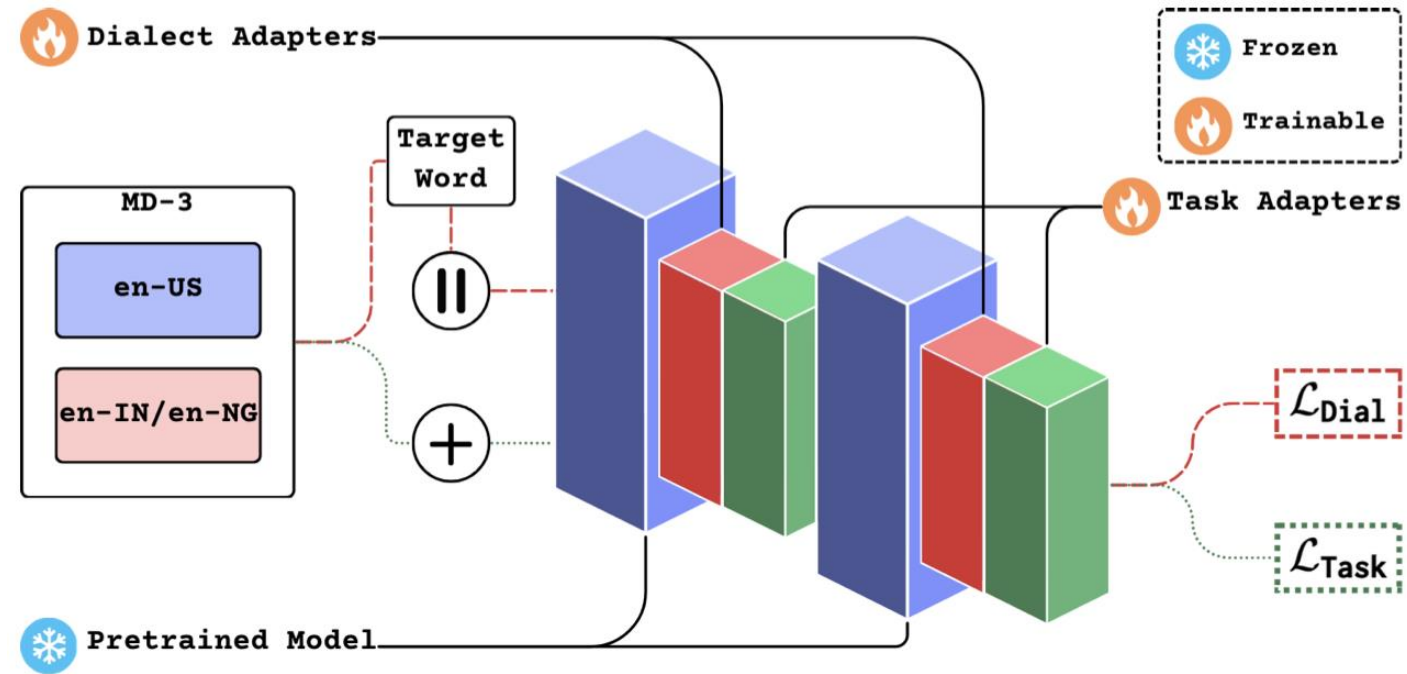
Indian English (en-IN)

Nigerian English (en-NG)

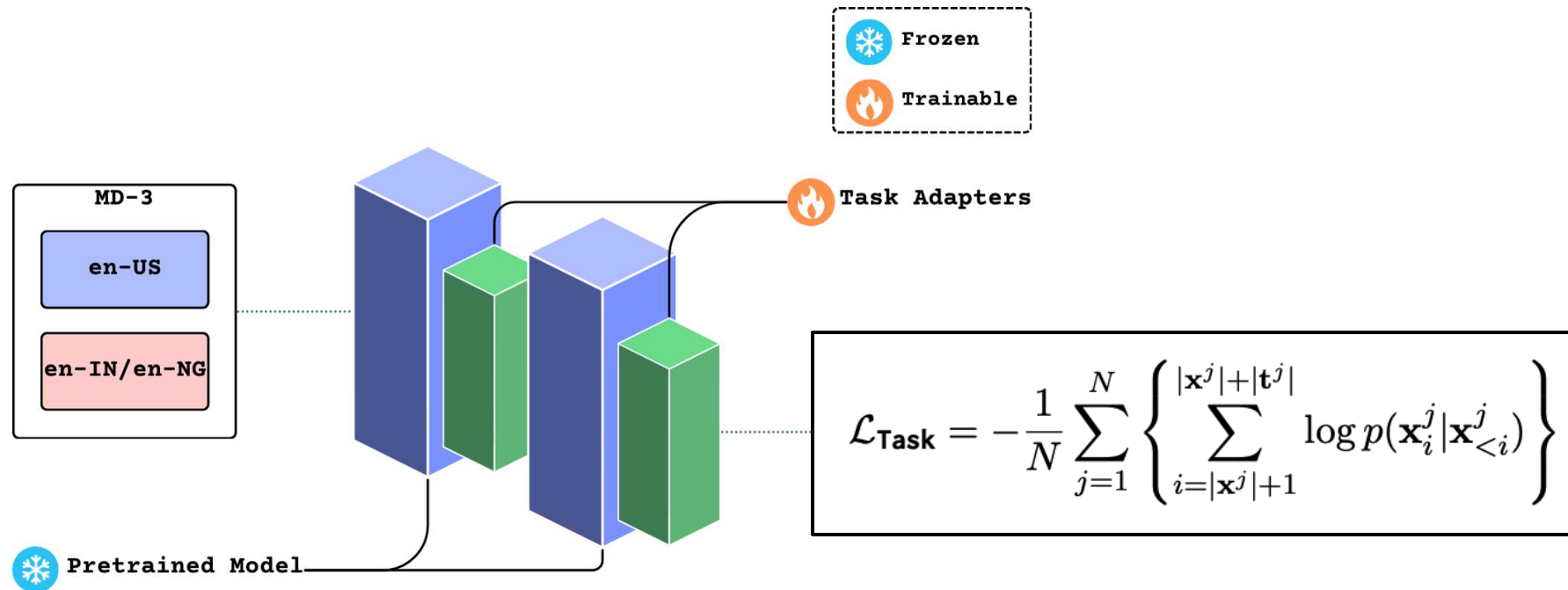


..come to the point!

We train dialect adapters using contrastive learning objective.

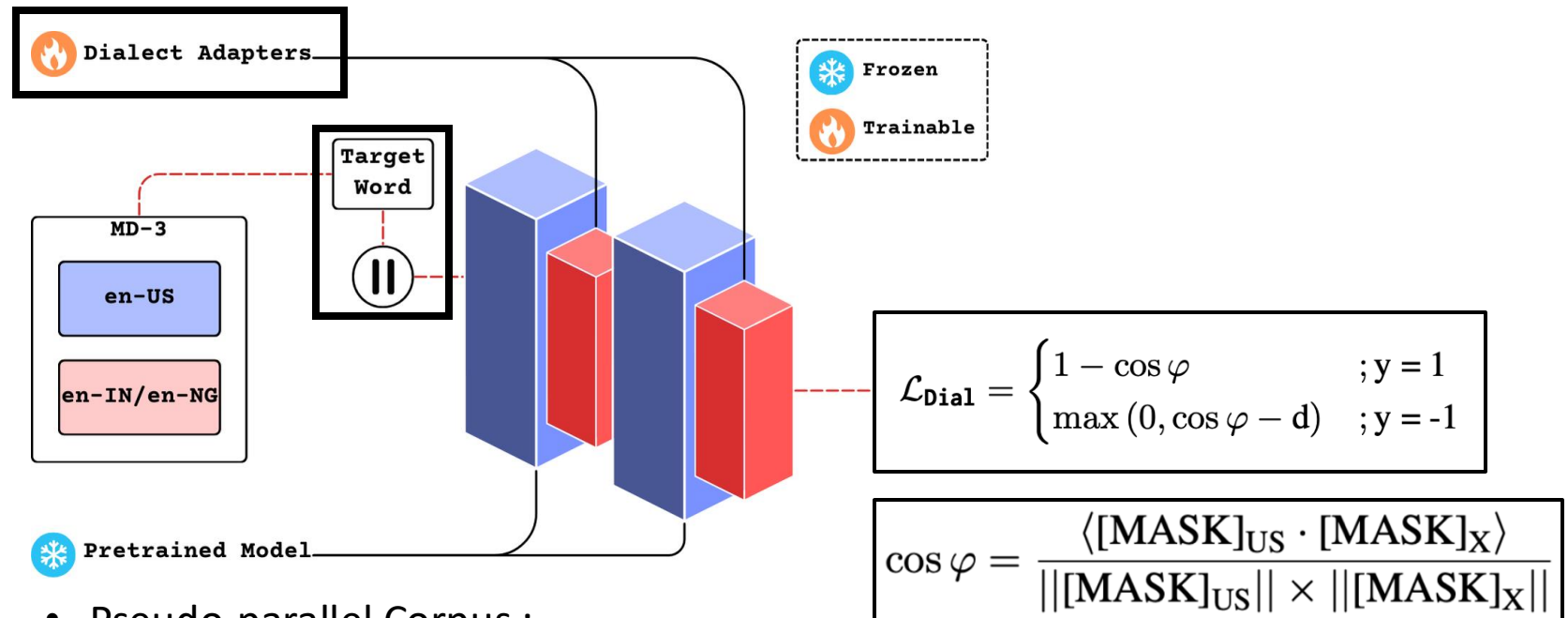


Step 1: Task Adapters



LoRA as the task adapter for Target Word Prediction

Step 2: Dialect Adapters



- Pseudo-parallel Corpus :

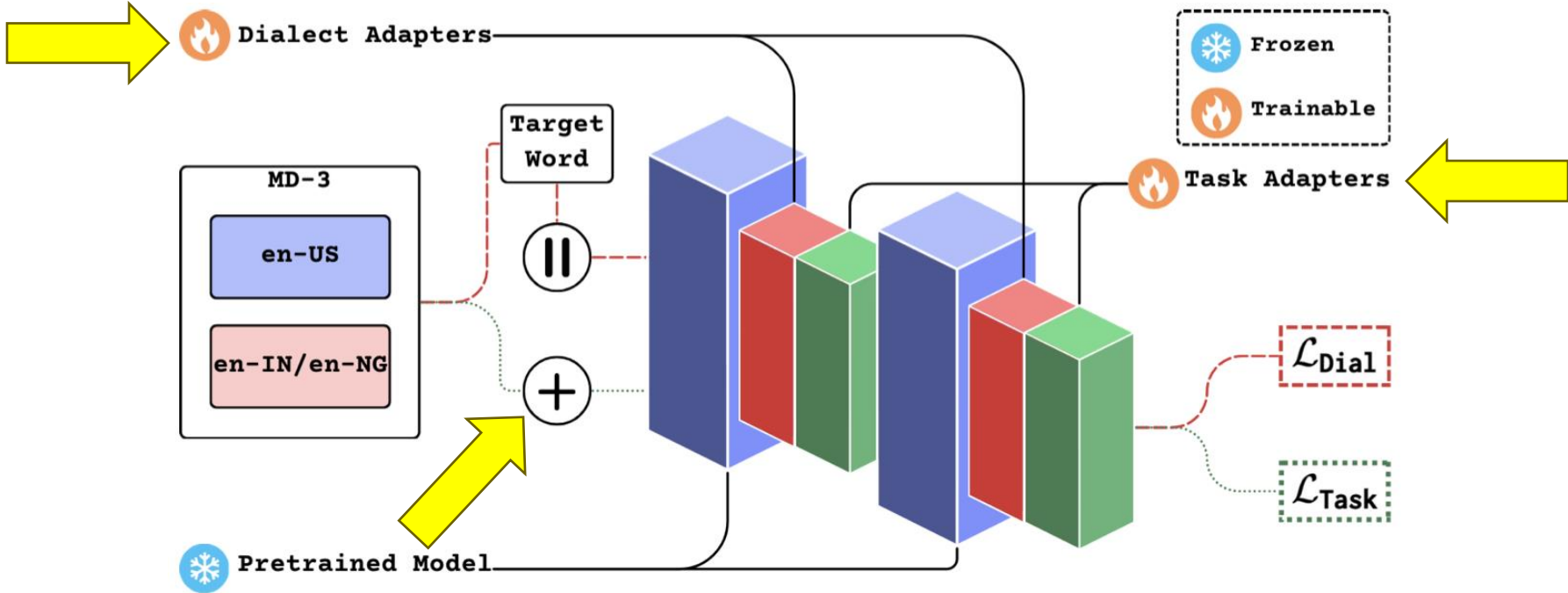
- en-US and corresponding en-IN/en-NG conversations for the same target word.
- With negative sampling (different target word).

- Contrastive Loss between:

- Frozen representation ($[\text{MASK}]_{\text{US}}$)
- Learnable representation ($[\text{MASK}]_{\text{X}}$)

LoRA as the dialect adapter

L o R D D

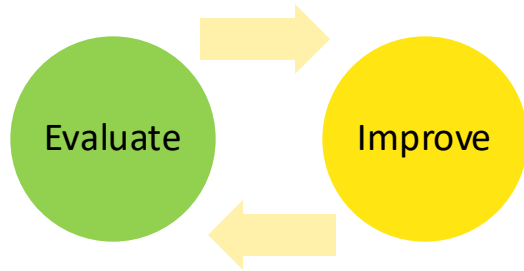


Low Rank Dialect Adapters for Decoder Models

M m m , o k a y ... s h o w m e t h e n u m b e r s !

Method	Training Data	MISTRAL		GEMMA		μ	
		Similarity	Accuracy	Similarity	Accuracy	Similarity	Accuracy
Skyline	en-US	64.7	44.3	69.7	45.0	67.2 (27.3)	(0.0) 44.8 (64.7)
(a) Tested on en-IN							
In-dialect baseline	en-IN	51.0	24.4	54.6	30.0	52.8 (0.0)	(64.7) 27.2 (0.0)
	en-US	54.6	25.6	61.3	35.0	58.0	30.3
Cross-dialect baseline	IN-MV	52.4	24.4	58.2	30.0	55.3	27.2
	IN-TR	50.4	24.3	53.0	26.9	52.7	25.6
LoRDD	en-US + en-IN	55.9	30.0	63.9	41.3	59.9 (13.4)	35.7 (28.1)
(b) Tested on en-NG							
In-dialect baseline	en-NG	53.0	27.2	60.9	35.3	(17.9) 57.0 (0.0)	(43.1) 31.3 (0.0)
	en-US	58.9	31.4	62.8	40.7	60.9	36.1
Cross-dialect baseline	NG-MV	55.7	28.4	61.4	38.0	58.9	33.5
LoRDD	en-US + en-NG	62.4	40.5	64.5	43.2	63.5 (11.4)	41.9 (33.8)

Part 3: Evaluate



BESSTIE: A Benchmark for Sentiment and Sarcasm Classification for Varieties of English

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Funded by



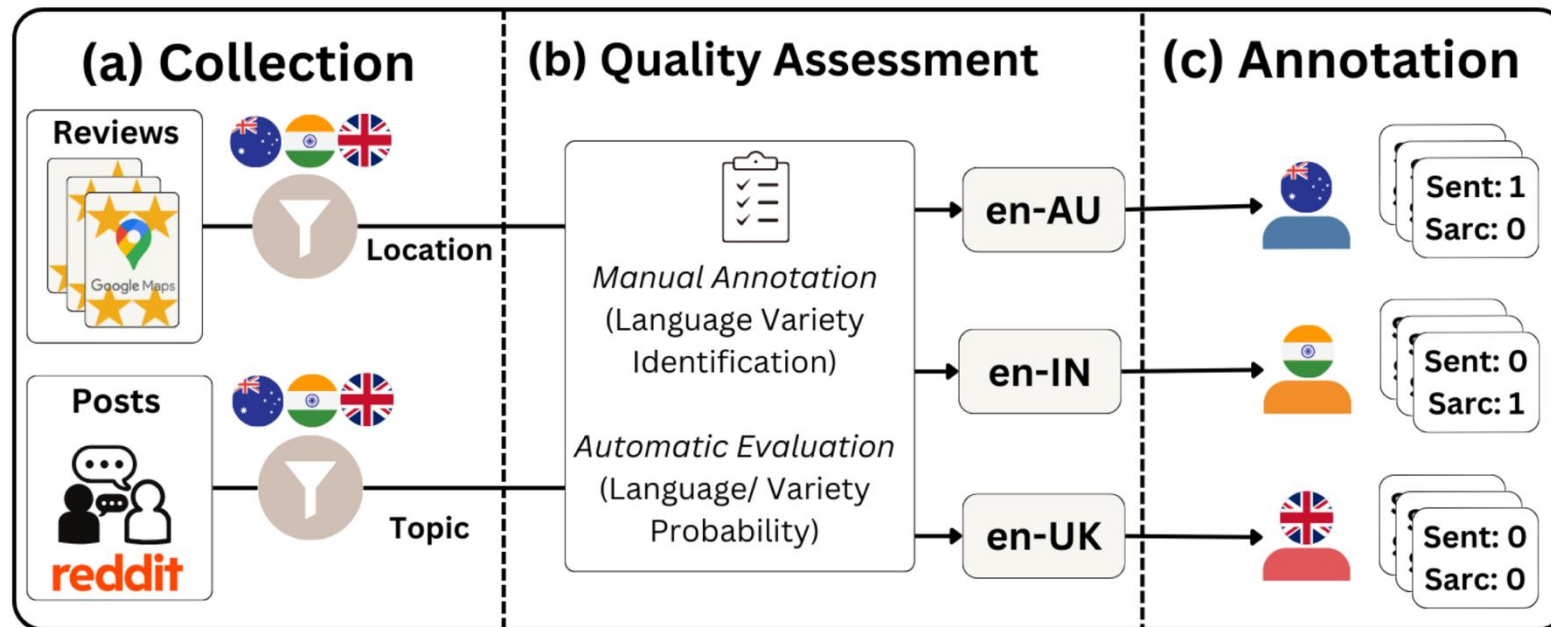
TL;DR: We have created a new dialectal benchmark for sentiment and sarcasm classification.

Let's handle the most popular downstream NLP task

Benchmark	Sent.	Sarc.	Eng.	Var.
Cieliebak et al. (2017) [6]	✓	×	×	×
Wang et al. (2018) [37]	✓	×	✓	×
Alharbi et al. (2020) [4]	✓	×	×	✓
Abu Farha et al. (2021) [3]	✓	✓	×	✓
Elmadany et al. (2023) [10]	✓	✓	×	✓
Faisal et al. (2024) [11]	✓	×	×	✓
BESSTIE	✓	✓	✓	✓

Table 1: Comparison of BESSTIE with past benchmarks for sentiment or sarcasm classification. ‘Sent.’ indicates sentiment classification, ‘Sarc.’ denotes sarcasm classification, ‘Eng.’ denotes English, and ‘Var.’ denotes language varieties. A checkmark (✓) denotes the availability of a particular feature, while a cross (×) indicates its absence.

Data Collection



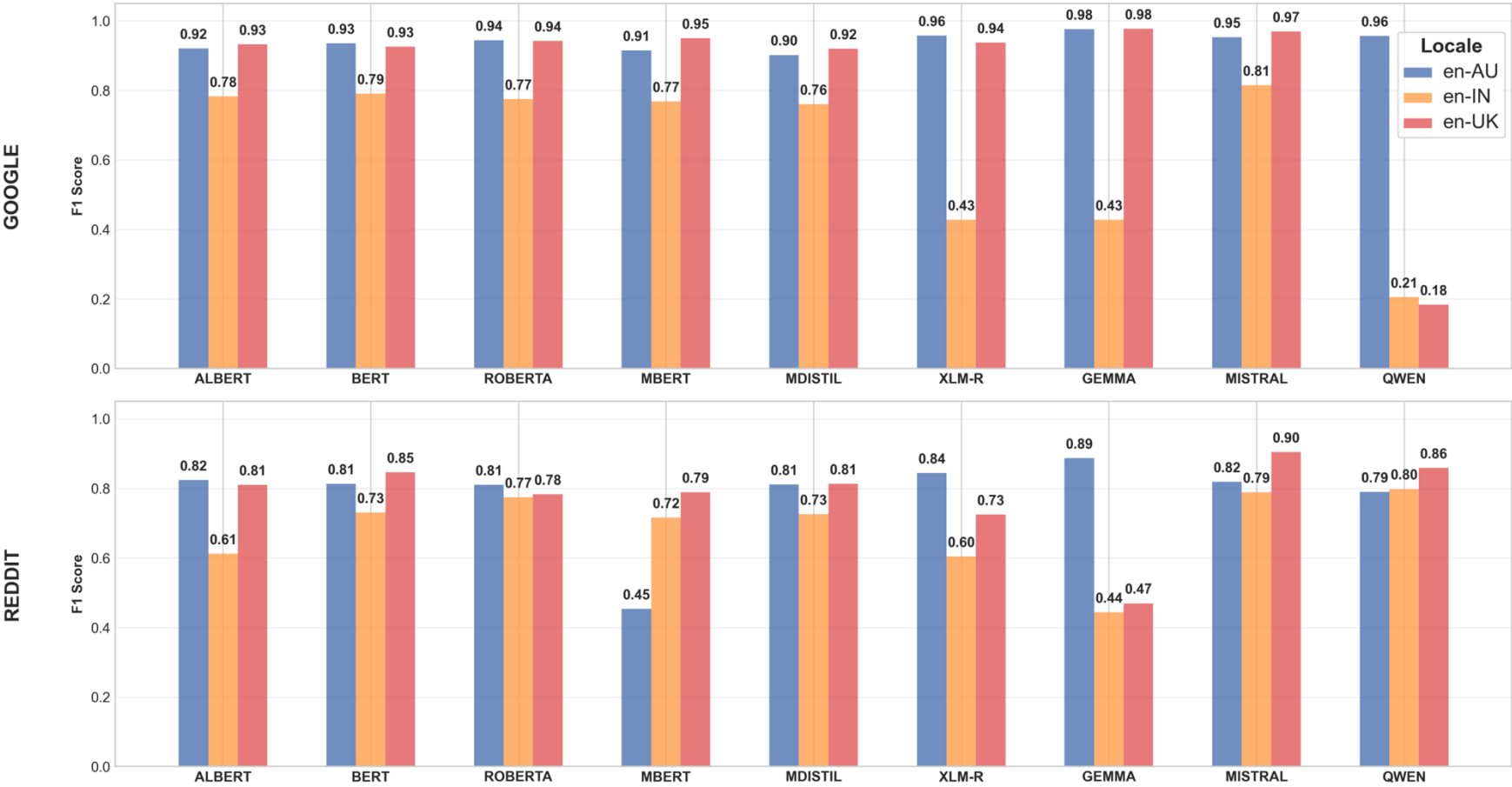
BESSTIE Dataset

Variety	Comment	Sentiment	Sarcasm
en-AU	Well set out, good stock and very friendly country staff	1	0
en-AU	S**t. Christ isn't back, is he?	0	1
en-IN	Good quality foods are supplied here	1	0
en-IN	Coz we all have free internet.	0	1
en-UK	Traditional friendly pub. Excellent beer	1	0
en-UK	What a brave potatriot	0	1

Table 2: Reviews and comments from different varieties with their corresponding annotated sentiment and sarcasm labels.

Variety	Subset	Train	Valid	Test	% Positive Sentiment	% Positive Sarcasm	$P(\text{eng})$	$\bar{w}$
en-AU	GOOGLE	946	130	270	73%	7%	0.99	63.97
	REDDIT	1763	241	501	32%	42%	0.98	51.72
en-IN	GOOGLE	1648	225	469	75%	1%	0.99	44.34
	REDDIT	1686	230	479	25%	13%	0.87	26.92
en-UK	GOOGLE	1817	248	517	75%	0%	0.99	72.21
	REDDIT	1007	138	287	12%	22%	0.98	38.04

Results: Sentiment Classification



Results: Sarcasm Classification

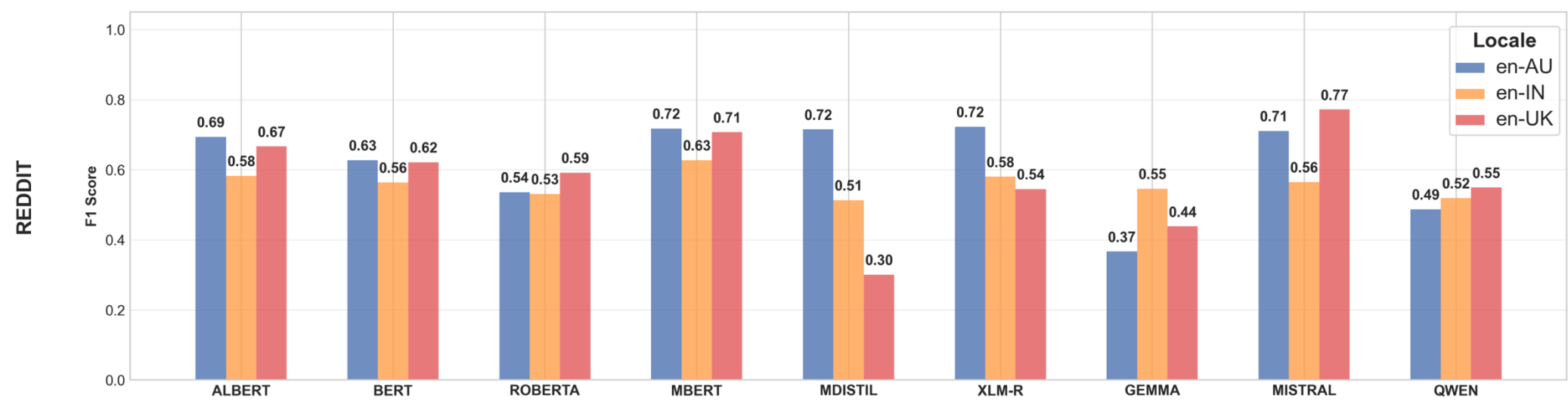


Figure 4: Performance comparison of various models on the sarcasm classification task across different English varieties (en-AU, en-IN, and en-UK).

Putting it all together..

Domain-Task	en-AU	en-IN	en-UK
GOOGLE-Sentiment	0.94	0.64	0.86
REDDIT-Sentiment	0.78	0.69	0.78
REDDIT-Sarcasm	0.62	0.56	0.58
μ	0.78	0.63	0.74

Table 4: Task performances, averaged across all models, on all varieties. μ represents the average performance across all domain-task pairs.

Model	Enco.	Mono.	GOOGLE-Sentiment	REDDIT-Sentiment	REDDIT-Sarcasm
ALBERT	✓	✓	0.88	0.75	0.65
BERT	✓	✓	0.88	0.80	0.60
ROBERTA	✓	✓	0.89	0.79	0.55
MBERT	✓	✗	0.88	0.65	0.68
MDISTIL	✓	✗	0.86	0.78	0.51
XLM-R	✓	✗	0.77	0.73	0.62
GEMMA	✗	✓	0.79	0.60	0.45
MISTRAL	✗	✗	0.91	0.84	0.68
QWEN	✗	✗	0.45	0.82	0.52
μ	$\frac{\checkmark: 0.74}{\times: 0.67}$	$\frac{\checkmark: 0.72}{\times: 0.71}$	0.81	0.75	0.59

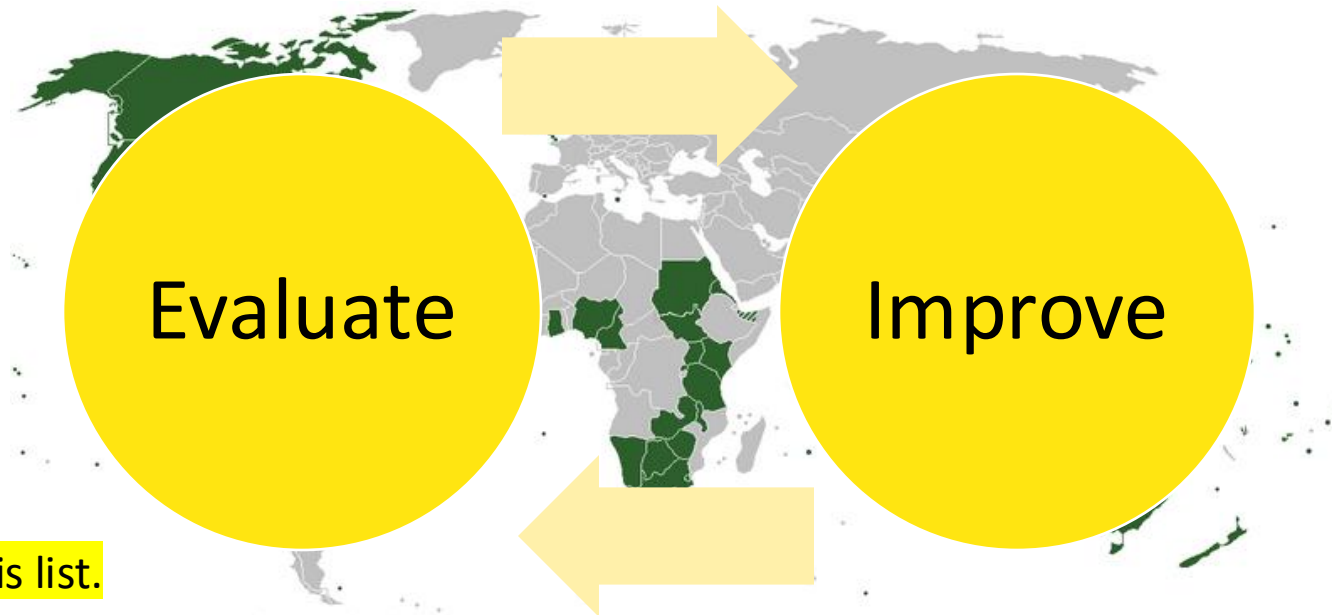
Summary

Evaluating the ability to predict the topic of conversations.

Evaluating the ability to detect sentiment and sarcasm.

We would like to grow this list.

Also see: Pragmatic Metacognitive Prompting for Explainable Sarcasm Classification: <https://arxiv.org/abs/2505.15095>



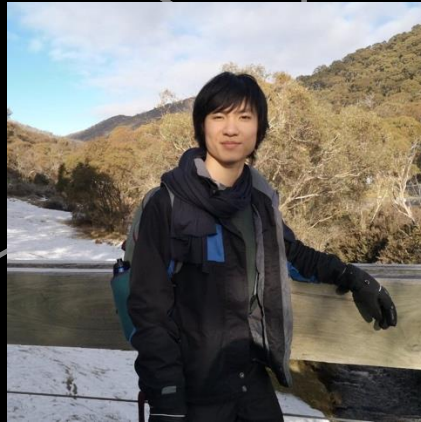
Enhancing the ability to predict the topic of conversations.

..and this one too.

Use-cases?! Users with heterogenous linguistic backgrounds.



Australia's
Global
University



Part 2:

Multilingual hate speech classification

Funded by Google Research as a Taste-of-Research Scholarship offered to high-performing undergraduate students at UNSW.

M o t i v a t i o n

Transformer -> Translation

Are nuances of hate speech preserved in translation though?

“Is Hate Lost in Translation?": Evaluation of Multilingual LGBTQIA+ Hate Speech Detection. Fai Leui Chan, Duke Nguyen, Aditya Joshi. ALTA 2024, Canberra, Australia.

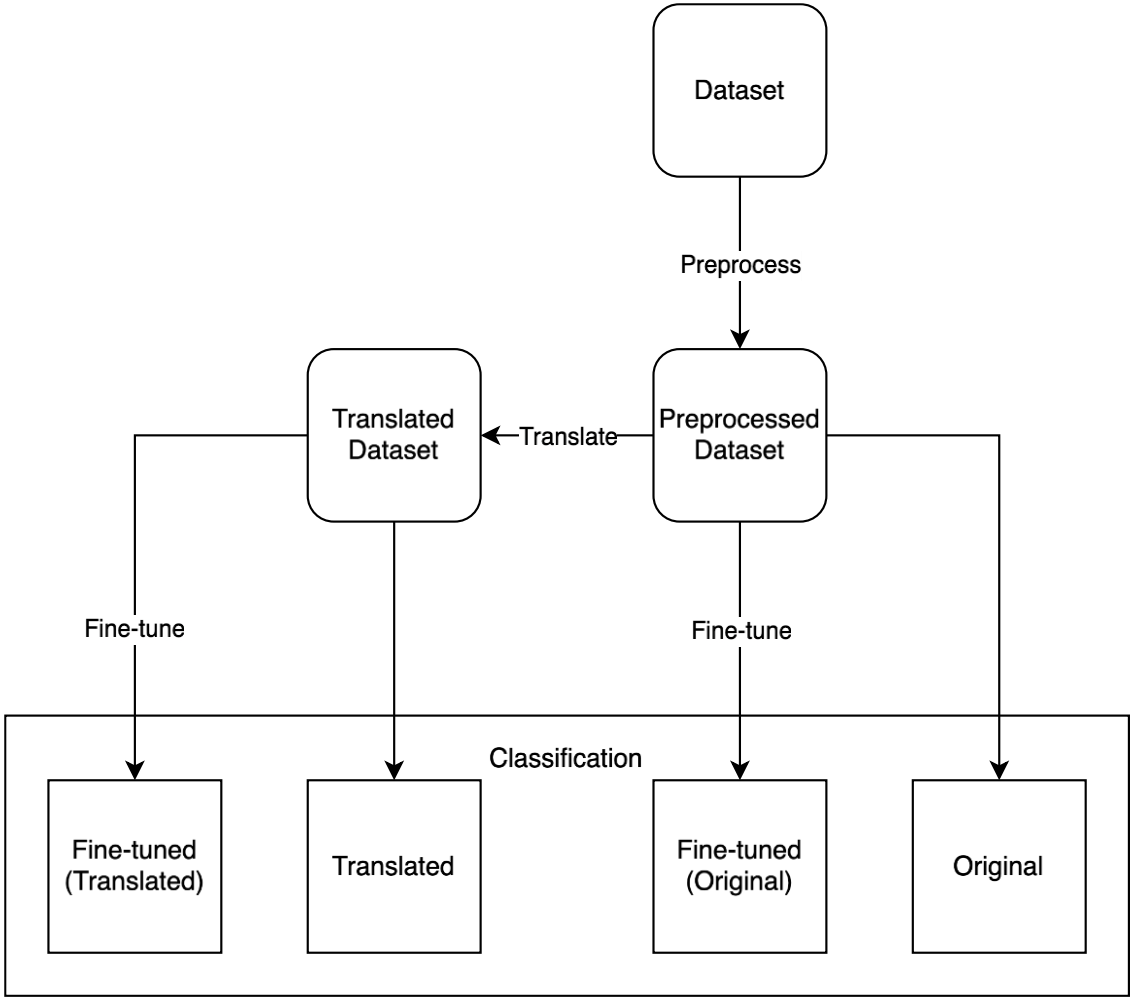
We evaluate for:

- English, Italian, Chinese and (code-switched) Tamil

- Hate speech detection

- ...towards LGBTQAI+ people

Methodology



Experiment setup

Language	Source	Total Samples	Non-Homotransphobic	Homotransphobic
English	(McGiff and Nikolov, 2024)	1,277	656 (51.4%)	621 (48.6%)
Italian	(Nozza et al., 2023)	5,000	2,992 (59.8%)	2,008 (40.2%)
Chinese	(Lu et al., 2023)	2011	1247 (62.0%)	764 (38.0%)
English-Tamil	(Chakravarthi et al., 2021)	6033	5384 (89.2%)	649 (10.8%)

Table 1: Comparison of datasets. % in the Non-Homotransphobic and Homotransphobic columns refer to the proportion of each class relative to the total samples in each dataset, with each row summing to 100%.

E x p e r i m e n t s e t u p

Model

GPT-3.5-turbo model

Metrics

F1 score

Precision

Recall

Cohen's Kappa

Results: quantitative analysis

Language	Condition	F1	P	R	K	ΔF	ΔK
English	Original	0.7952	0.7082	0.9066	0.5488	-	-
	Fine-tuned	0.8689	0.8833	0.8548	0.7486	+0.0737	+0.1998
Italian	Original	0.5990	0.4514	0.8899	0.1414	-	-
	Translated	0.5355	0.4424	0.6783	0.0960	-0.0635	-0.0454
	Fine-tuned (Original)	0.8375	0.8417	0.8333	0.7292	+0.2385	+0.5878
	Fine-tuned (Translated)	0.7417	0.7371	0.7463	0.5662	+0.2062	+0.4702
Chinese	Original	0.7464	0.7493	0.7435	0.5878	-	-
	Translated	0.6839	0.7099	0.6597	0.2463	-0.0625	-0.3415
	Fine-tuned (Original)	0.8146	0.8255	0.8039	0.7030	+0.0682	+0.1152
	Fine-tuned (Translated)	0.7661	0.7958	0.7386	0.6308	+0.0822	+0.3845
English- Tamil	Original	0.3619	0.2843	0.4977	0.1998	-	-
	Translated	0.3202	0.3511	0.2943	0.2463	-0.0417	+0.0465
	Fine-tuned (Original)	0.5391	0.6200	0.4769	0.2452	+0.1772	+0.0454
	Fine-tuned (Translated)	0.4037	0.5000	0.3385	0.3469	+0.0835	+0.1006

Table 2: Performance Metrics (F1: F1-score, P: Precision, R: Recall: K: Cohen’s Kappa) and Changes Across Languages and Conditions. All scores are weighted. Δ columns represent the changes in F1-score and Cohen’s Kappa between different conditions: Fine-tuned (Original $\rightarrow$ Fine-tuned), Translated (Original $\rightarrow$ Translated), and Fine-tuned (Translated $\rightarrow$ Fine-tuned).

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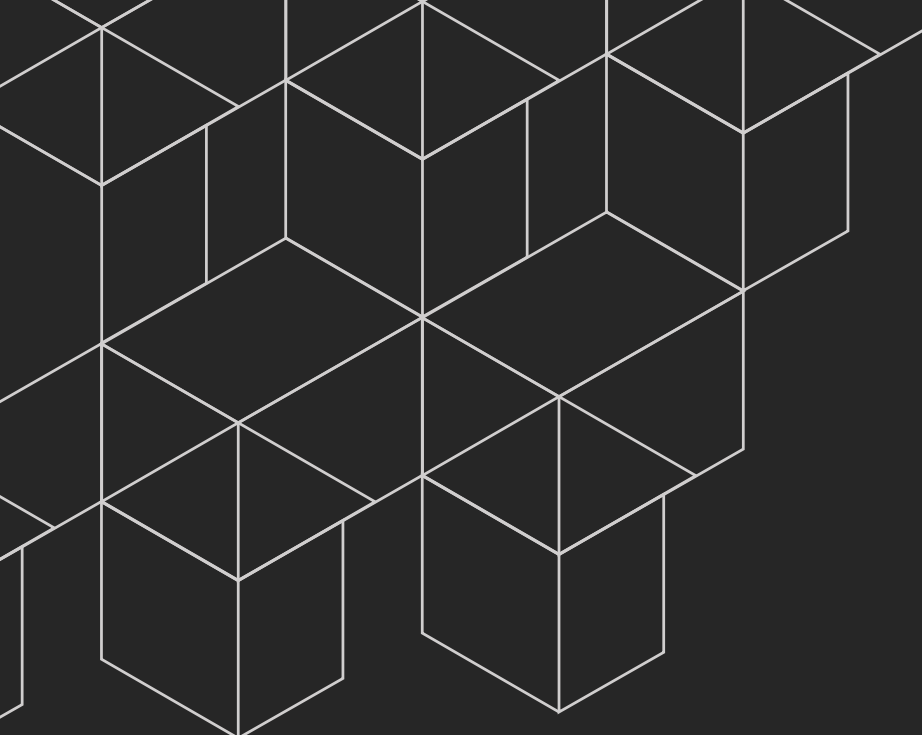
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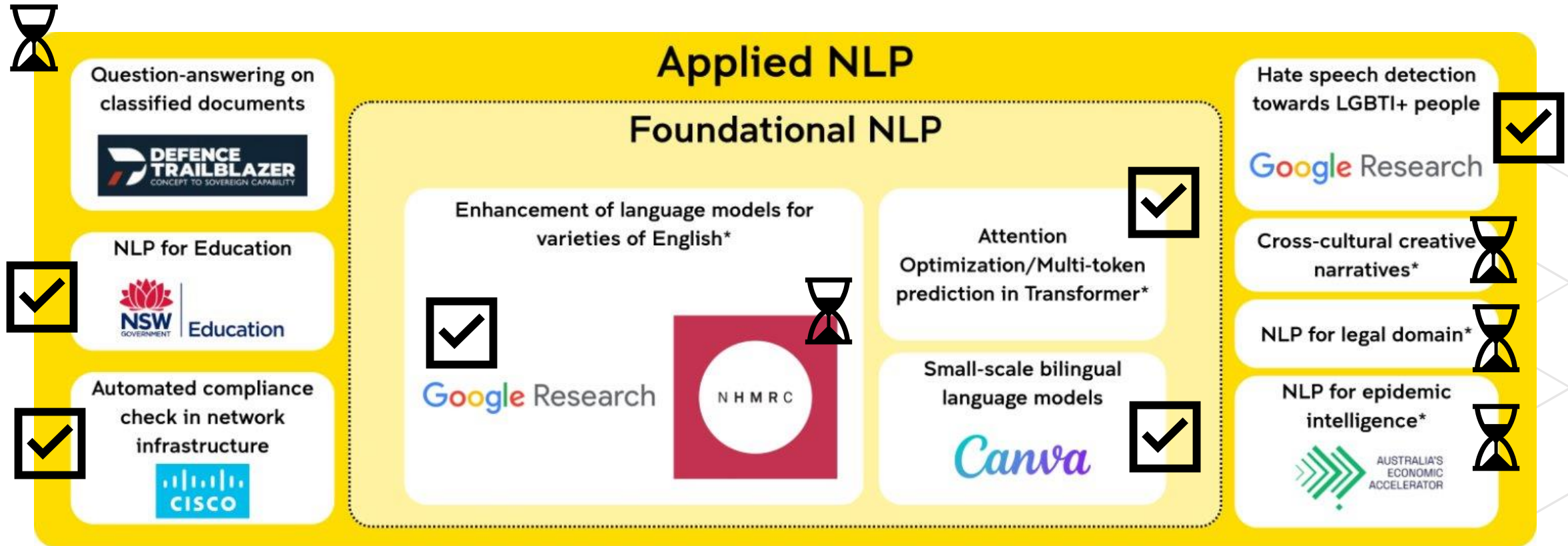
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Wrapping up...

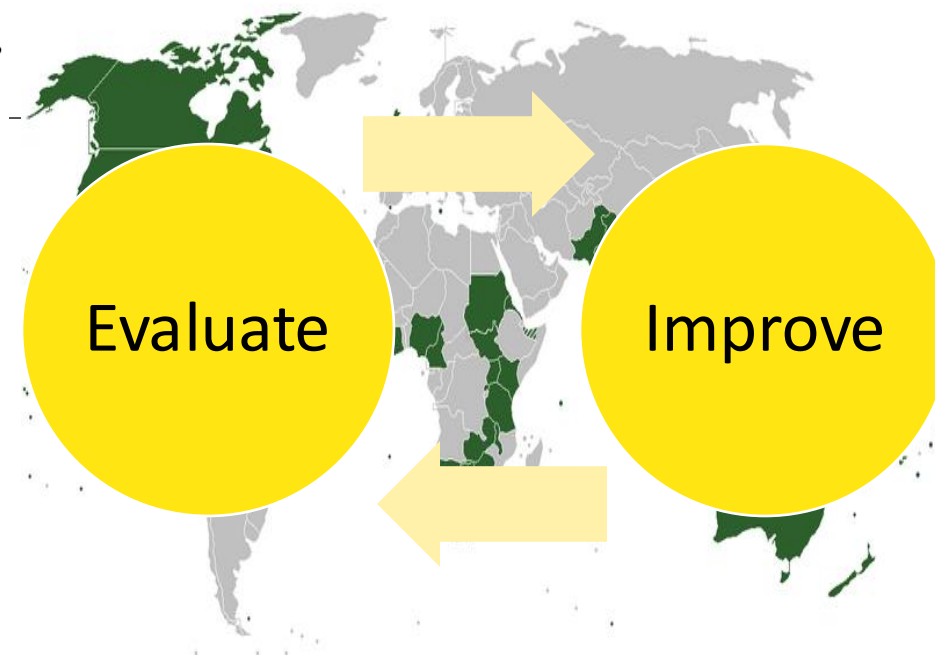
NLP Research at UNSW



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Table 4: Task performances, averaged across all models, on all varieties. μ represents the average performance across all domain-task pairs.

thank you!



Aditya Joshi | aditya.joshi@unsw.edu.au

